Analysis of Pediatric Ventricular Populations of Different Age Ranges

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Abstract

Pediatric cardiomyocytes exhibit electrophysiological features that differ markedly from those of adults, reflecting developmental changes in ionic currents, calcium handling, and action potential morphology [4-6]. Despite the widespread use of adult-based ventricular models, quantitative frameworks capable of capturing age-specific electrophysiological variability during postnatal development remain limited [1]. In this work, we generated ventricular model populations for five pediatric age groups (Neonate, Lactante, Niño pre-escolar, Edad escolar, Adolescente) using the O'Hara–Rudy model adapted with age-dependent ionic conductances [1]. Each population was simulated at 0.5 and 1 Hz, and four biomarkers—APD90, CaTD90, triangulation, and ICaL_peak—were extracted to assess both electrical and calcium-related properties.

The results reveal a clear maturational progression across age groups: younger models display longer action potentials and calcium transients, increased action potential triangulation, and a stronger influence of ICaL [6], whereas older groups exhibit more efficient repolarization and faster calcium reuptake [4,5]. An increase in pacing frequency produced a consistent shortening of all biomarkers, in agreement with known frequency-dependent electrophysiological behavior [1].

1. Introduction

Cardiac electrophysiological maturation from birth to adolescence involves progressive changes in ionic currents, calcium handling, and action potential duration [4-6]. Foundational studies on L-type calcium current [6], computational ventricular models [1], and recent pediatric transcriptomic analyses [4,5] consistently demonstrate marked age-dependent differences in these mechanisms.

Although robust adult models such as O'Hara–Rudy are widely used [1], relatively few studies have quantified how physiological variations in ionic conductances (±15–30%) affect electrophysiology across pediatric age ranges [3]. Development is known to alter repolarization reserve and calcium dynamics [4-6], yet only limited work has evaluated these effects using age-specific models that integrate action potential morphology, calcium transients, and sensitivity analysis.

Computational modelling provides a powerful framework to study such variability. Population-based approaches, as described by Llopis-Lorente et al. [2,3], allow systematic exploration of how changes in ionic properties influence biomarkers like APD90, CaTD90, triangulation, and ICaL_peak.

Based on this evidence, the aim of this work is to characterize the evolution of key electrophysiological biomarkers across five pediatric stages (Neonate, Lactante, Niño pre-escolar, Edad escolar, Adolescente). Age-specific ventricular populations composed exclusively of endocardial-type cells were simulated at 0.5 and 1 Hz, and the analysis focused on APD90, CaTD90, triangulation, and ICaL_peak to assess how electrophysiological properties and their rate dependence change during postnatal maturation.

2. Methods

2.1. Electrophysiological model

Simulations were based on the human ventricular O'Hara–Rudy (ORd, 2011) model, one of the most reliable references for quantitatively describing the action potential and ionic currents in adult cardiomyocytes [1]. The model reproduces in detail the major depolarizing and repolarizing currents, as well as intracellular calcium dynamics [1].

Starting from this adult formulation, five age-specific model variants were implemented by selectively adjusting ionic conductances according to the values reported in the pediatric transcriptomic analyses [4,5].

2.2 Pediatric populations and parameter variation

Five ventricular populations were generated, corresponding to different age-specific models. For each age group, 12 key conductances identified as age-dependent were varied individually (one-at-a-time) at five scaling levels (−30%, −15%, nominal, +15%, +30%), producing 60 simulations per age and per pacing frequency. With two pacing frequencies (0.5 and 1 Hz), the full dataset consisted of 600 simulations. Pacing frequency was not analysed as an independent factor; its sole purpose was to verify the robustness of biomarker responses. This approach follows the population-of-models framework of Llopis-Lorente et al. [2,3], enabling systematic exploration of physiological variability.

2.3 Simulations and extraced biomakers

Simulations were run to steady state, and four biomarkers were extracted: APD90 (action potential duration at 90% repolarization), CaTD90 (calcium transient duration), Triangulation (APD90 − APD50), ICaL_peak (peak L-type calcium current) [1,6].
The analysis focused primarily on APD90 and CaTD90, which are widely used indicators of electrophysiological and calcium-handling maturation [4,5]. Final beats were stored in MATLAB *.mat* format for visualization.

2.4 Data analysis and visualization

Variability and age-dependent trends were assessed using boxplots of APD90 and CaTD90. Superimposed action potential and Ca^{2+} traces provided qualitative comparison across ages. Relative sensitivity heatmaps quantified the influence of ionic parameters on each biomarker for all age groups. Additional exploratory boxplots were generated to identify influential parameters but were not included in the final set of results.

3. Results

3.1 Distribution of APD90 and CaTD90 across age groups

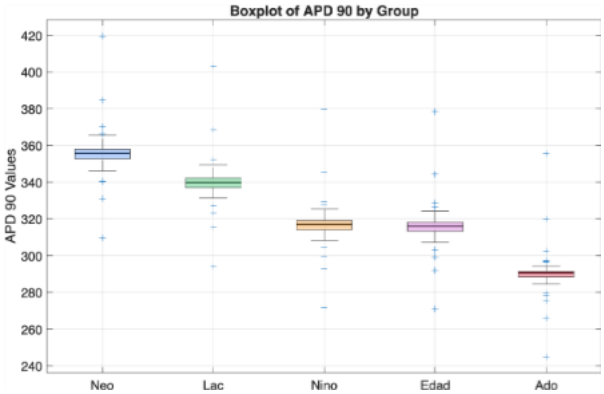


Figure 1. the APD90 boxplots for all age groups under the most influential scaling condition with multiplier 0.7 on GKr parameter.

Based on preliminary sensitivity analyses across all age groups (not shown), GKr was identified as the parameter exerting the strongest influence on APD90 and CaTD90 and was therefore selected for illustrative comparison.

The analysis of the five age-specific models enabled a systematic characterization of how physiological variations in ionic conductances affect the main electrophysiological biomarkers. For each age group, 60 simulations were performed by individually varying the twelve conductances at −30%, −15%, nominal, +15%, and +30%.

Only the results at 1 Hz are presented here, as this pacing frequency serves as the reference condition. Simulations at 0.5 Hz were used solely to verify restitution consistency and are briefly discussed in Section 3.4.

Figure 1 shows the distribution of APD90 for the five age groups. A progressive reduction in APD90 is observed from the neonatal to the adolescent model, accompanied by a marked decrease in dispersion. Neonates exhibit a longer and more variable repolarization phase, whereas in more mature models repolarization becomes increasingly stable and compact. This trend reflects the increasing contribution of repolarizing currents during development, in agreement with experimental evidence on postnatal ionic maturation [4,5].

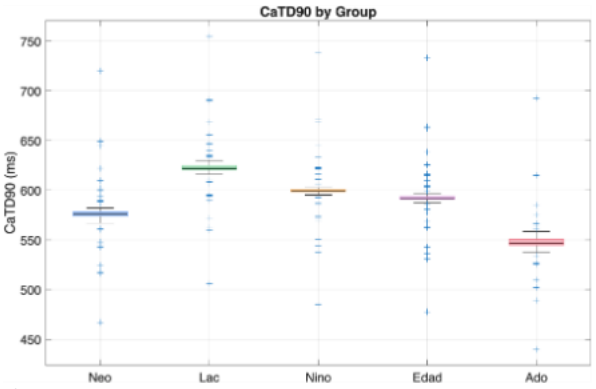


Figure 2. The CaTD90 boxplots for all age groups under the most influential scaling condition with multiplier 0.7 on GKr parameter.

A similar pattern emerges in Figure 2, which reports the distribution of CaTD90. Younger models display higher values and greater inter-simulation variability, indicating an immature calcium-reuptake system. With increasing age, the calcium transient duration decreases progressively and becomes shorter and more uniform in adolescents, consistent with enhanced SERCA and NCX function, as also supported by the sensitivity analysis presented in Section 3.3 [4,5].

3.2 Qualitative analysis of action potentials and calcium transients

To provide a more intuitive qualitative overview, Figures 3 and 4 show representative action potentials and calcium transients for the five age groups. In the action potential (Figure 3), the plateau phase clearly shortens, and repolarization becomes faster with age. Similarly, the calcium transients (Figure 4) exhibit a slower decay and prolonged duration in neonates, whereas more mature models show a sharper peak and a faster return to baseline. Together, these profiles visually illustrate the

progressive electrophysiological maturation across developmental stages.

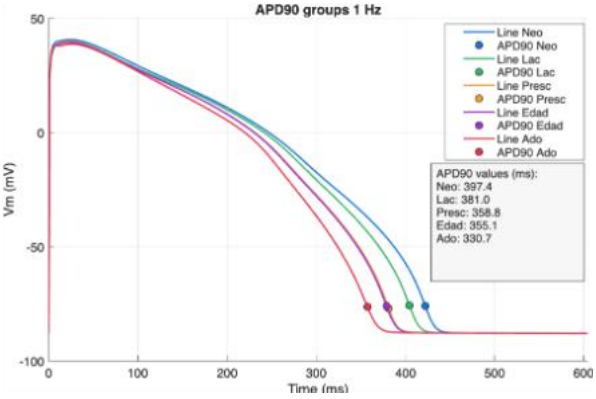


Figure 3. Representation of APD90 across age groups under the strongest parameter change (scaling factor 0.7 on GKr parameter).

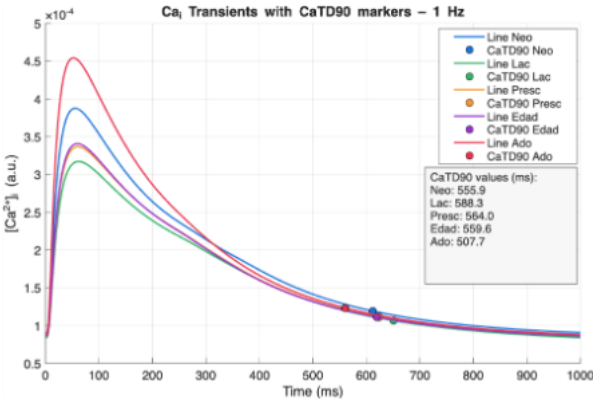


Figure 4. Representation of CaTD90 across age groups under the strongest parameter change (scaling factor 0.7).

3.3 Sensitivity analysis

To mechanistically interpret the trends observed in Sections 3.1 and 3.2, a normalized sensitivity heatmap was constructed (Figure 5). In the neonatal model, the dominant role of GKr is evident, emerging as the main modulator of APD90 and triangulation. The late sodium current (GNaL) shows a particularly strong influence during early development: even modest changes in this parameter can substantially affect triangulation and calcium-transient duration. Regarding calcium dynamics, sensitivity is dominated by SERCA (GSERCA) and sarcoplasmic-reticulum release (Jrel), confirming their central contribution to the maturation of intracellular calcium handling.

Overall, the 1 Hz sensitivity analysis reveals a coherent picture: younger age groups are more susceptible to parameter perturbations, whereas older models appear more stable due to more efficient repolarization and calcium-handling mechanisms.

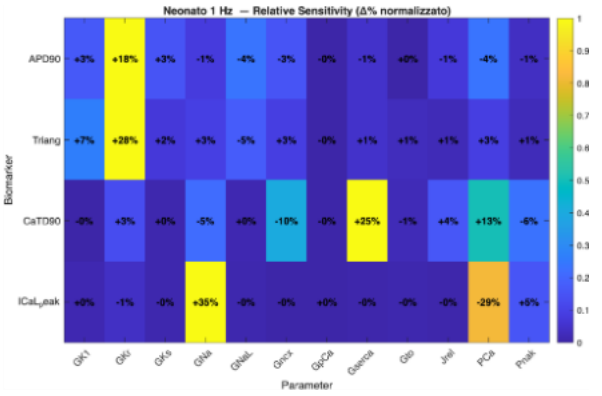


Figure 5. Relative sensitivity heatmap for the Neonate model across all biomarkers and ionic parameters.

3.4 Verification at 0.5 Hz

For completeness, additional simulations were performed at 0.5 Hz (not shown). These simulations exhibited behaviour fully consistent with the 1 Hz results: The restitution relationship between diastolic interval and action potential duration was preserved for all ages, the hierarchy of influential parameters remained unchanged, no reversals or unexpected phenomena were observed.

This confirms that the conclusions regarding the influence of conductances on electrophysiological biomarkers are robust and independent of pacing frequency.

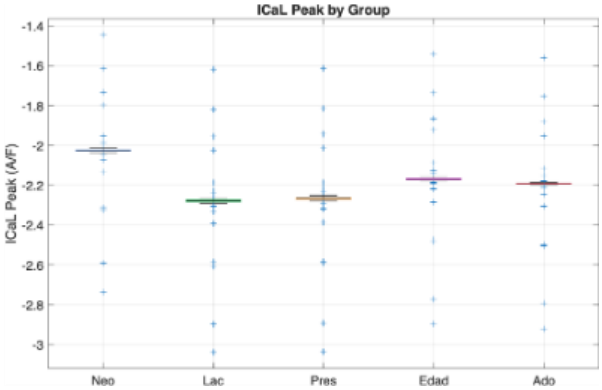


Figure 6. ICaL peak boxplots for all age groups under the most influential scaling condition (multiplier 0.7 on GKr parameter).

4. Discussion

The results demonstrate that pediatric electrophysiological maturation affects both the baseline values of key biomarkers and their sensitivity to ionic conductance variations. A clear developmental progression emerges, consistent with literature describing the evolution of calcium handling and

repolarization mechanisms during postnatal growth [4-6].

APD90 decreases with age, together with reduced variability, reflecting the maturation of repolarizing potassium currents, particularly IK1 and Ikr [4,5], which strengthen the repolarization reserve in older models. Similarly, changes in CaTD90 highlight the transition from an immature calcium-handling system in neonates, characterized by slower decay and greater dispersion, to more efficient SERCA, and NCX, mediated calcium reuptake in later stages [4-6].

The sensitivity analysis further clarifies the ionic determinants of these trends. GKr consistently emerges as the main modulator of APD90 and triangulation, especially in neonatal models, confirming the vulnerability of early repolarization reserve [4,5]. GNaL also exerts a marked influence at younger ages, where small increases prolong depolarization and amplify triangulation, consistent with the less stable plateau observed in immature cardiomyocytes [4-6].

This study has some limitations. Age-specific models were obtained by scaling ionic conductances of an adult ventricular formulation rather than using pediatric-specific kinetics, and parameter variations were explored using a one-at-a-time approach. Moreover, simulations were performed at the single-cell level and do not account for tissue-scale interactions. Despite these limitations, the consistent age-dependent trends observed across biomarkers and pacing rates support the robustness of the proposed modeling framework.

4.1 Conclusions

Overall, the results show that early developmental stages exhibit greater sensitivity to changes in ionic conductances, reflected in both electrical behavior and calcium dynamics. This susceptibility decreases with age as repolarization mechanisms and calcium-handling pathways become more efficient. These findings are consistent with pediatric literature describing immature myocardium as having a reduced repolarization reserve and less stable calcium control, features that may underlie the higher arrhythmic vulnerability observed in very young patients [4-6]. The study also demonstrates that age-specific models qualitatively reproduce trends expected from experimental pediatric conductance data. The parameters exerting the strongest influence on the biomarkers correspond to those known to vary with development, supporting the validity of the modeling approach [4,5]. Overall, this highlights the value of computational methods for interpreting molecular and transcriptomic changes and for quantitatively characterizing electrophysiological maturation from birth through adolescence.

References

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